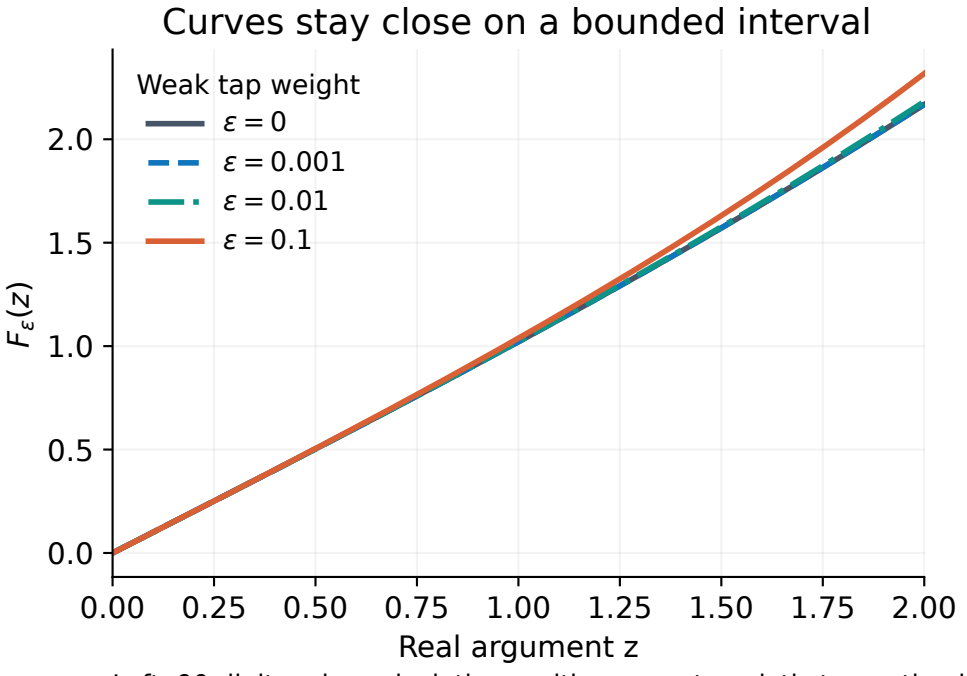
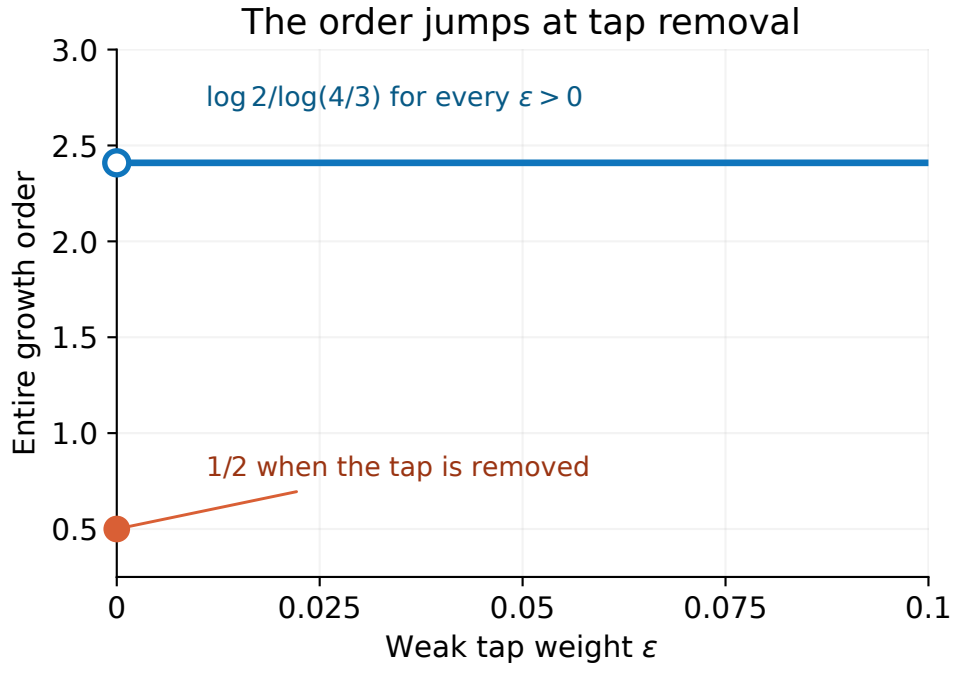
A weak feedback tap changes global growth

$$F_\varepsilon' = 1 + (1-\varepsilon)F_\varepsilon(z/4)^2 + \varepsilon F_\varepsilon(3z/4)^2, \quad F_\varepsilon(0) = 0, \quad 0 \le \varepsilon \le 1$$





Left: 90-digit series calculations, with an exact analytic truncation bound.

Right: the proved growth order. Compact convergence does not preserve this global invariant.